
Custodial Symmetry in Our RS Scan

What It Touches, Paper vs. Ours, and the $SU(2)_R$ / P_{LR} Split

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The one-line answer. Custodial symmetry is an *electroweak* cure, not a *flavor* cure. It protects the oblique T parameter and the $Z b_L \bar{b}_L$ vertex, pulling those floors from tens of TeV down to a few TeV, but it leaves the ε_K wall flat at tens of TeV, because the binding kaon operator is the color-octet KK-gluon C_4 , an electroweak singlet the custodial group cannot touch. This note works through that observable by observable, separates the two protections custodial actually provides, and lists which papers we followed.

1. What custodial symmetry touches, and how we implemented it

Custodial RS gauges $SU(2)_L \times SU(2)_R \times U(1)_X$ with the discrete P_{LR} . It is an electroweak symmetry: it acts on the oblique parameters and the $Z b \bar{b}$ vertex, and is blind to the color-octet KK-gluon operator that sets ε_K .

Observable	Touched?	Mechanism	How we implemented it
S, T, U oblique	yes (T)	$SU(2)_R$ removes the log L enhancement of T (ADMS); $\Delta T \propto +L \rightarrow -1/L$	custodial $\Delta T = -\frac{\pi v^2}{4c_W^2 M_{KK}^2 L}$ (CGHNP Eq. 153), <code>oblique_stu.py</code> ; S stays unprotected
$Z \rightarrow b \bar{b}, g_L^b$	yes (tree)	P_{LR} with $b_L \in (2, 2)_{2/3}$ zeroes δg_L^b at tree level (ACDP)	$\delta g_L^b \rightarrow 0$ on the protected mask (ACDP Eqs. 3–12); $1/L$ residual is a proxy knob
$Z \rightarrow b \bar{b}, g_R^b$	yes (zero)	$b_R \in (1, 1)_{-1/3}$: $T_L^3 = T_R^3 = 0 \Rightarrow \delta g_R^b = 0$ by P_C (ACDP Tab. 1)	$g_R^b = 0$ (singlet embedding, ACDP Tab. 1); not pinned in code
$Z \rightarrow b \bar{b}$ loops	partial	top partners re-introduce $-\delta g_L^b$, $+\Delta T(\text{singlet}) / -\Delta T(\text{bidoublet})$ (CPSW)	CPSW Eqs. 13–15, 35; ξ, ρ mass inputs are proxy knobs
$\varepsilon_K, \Delta m_K, \Delta F=2$	no	the KK-gluon C_4 operator is a color octet, electroweak <i>singlet</i> — unchanged under the custodial group	unchanged: <code>deltaf2.py</code> has no <code>ew_model</code> branch. Blanke’s full custodial calc confirms the wall

Our custodial run is a leading-order tree EW proxy; its four documented approximations and the full equation-by-equation provenance are in `docs/CUSTODIAL_PROVENANCE.md`, and the per-paper equation/figure map is in `docs/CUSTODIAL_PAPER_MAP.pdf`.

2. The two protections: $SU(2)_R$ protects T , P_{LR} protects Zb_L

Custodial bundles two distinct protections that do different jobs, and we can switch them on separately. The gauge $SU(2)_R$ tames the oblique T parameter; the extra discrete P_{LR} protects the Zb_L vertex. Running all three models (minimal, $SU(2)_R$ only, $SU(2)_R + P_{LR}$) gives the clean decomposition below: the oblique floor is the largest M_{KK} excluded at 95% against the PDG-2025 U -fixed ellipse, and δg_L^b is the Zb_L vertex shift at $M_{KK} = 3$ TeV.

EW model (code)	oblique S, T, U floor	Zb_L shift δg_L^b
minimal RS <code>minimal_rs</code>	16.0 TeV (T problem)	2.6×10^{-3} (unprotected)
$SU(2)_R$ only <code>custodial_rs_su2r</code>	5.8 TeV (T fixed; set by S)	2.6×10^{-3} (= minimal, still unprotected)
$SU(2)_R + P_{LR}$ <code>custodial_rs_plr</code>	5.8 TeV (unchanged)	0 (protected)

$SU(2)_R$ alone does the entire oblique job: it pulls the S, T, U floor from 16 down to 5.8 TeV (the leftover is the unprotected S), and it leaves Zb_L exactly where minimal had it. P_{LR} then does the Zb_L job: it collapses δg_L^b to zero without touching the oblique floor at all. So each symmetry cures one observable, independently. This is why the $Z \rightarrow b\bar{b}$ “no custodial” plot below stands for *both* minimal and $SU(2)_R$ -only: $SU(2)_R$ does not move those couplings, only P_{LR} does.

3. Paper vs. ours, constraint by constraint

In every comparison the published figure is on the left and our reproduction is on the right, in the same axes.

(a) S, T, U oblique — $SU(2)_R$ protects T

CGHNP Fig. 4 is one figure with a minimal (left) and a custodial (right) panel; P_{LR} does not change either.

No custodial

paper (left) | ours (right)

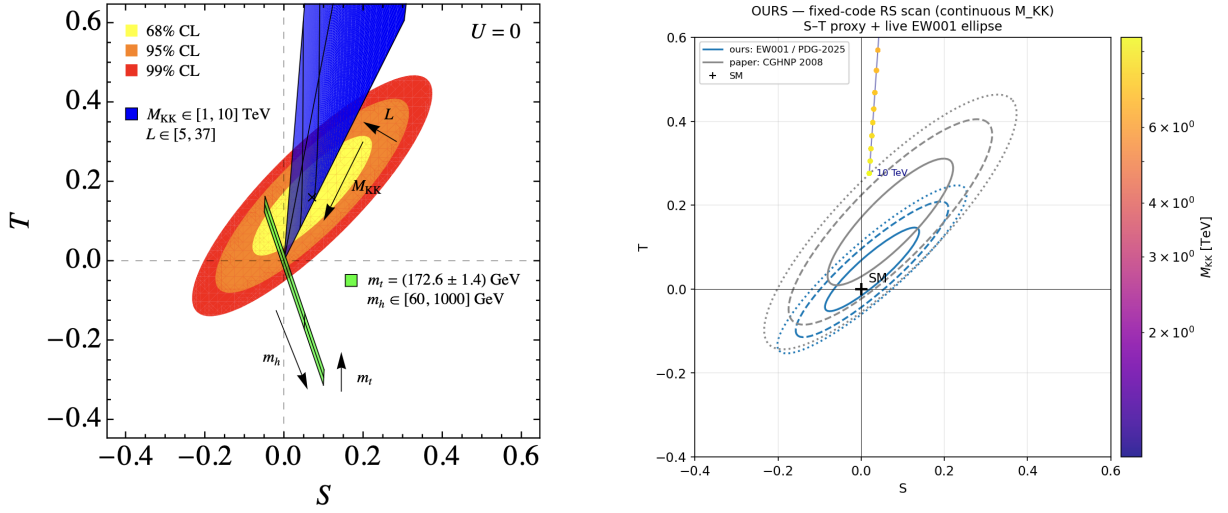


Figure 1: The S - T plane, same axes both sides. Minimal RS sweeps a wedge that climbs steeply in T (the volume-enhanced “RS T problem”) as M_{KK} runs $10 \rightarrow 1$ TeV against the CL ellipses. Our scan reproduces the paper wedge.

With custodial ($SU(2)_R$)

paper (left) | ours (right)

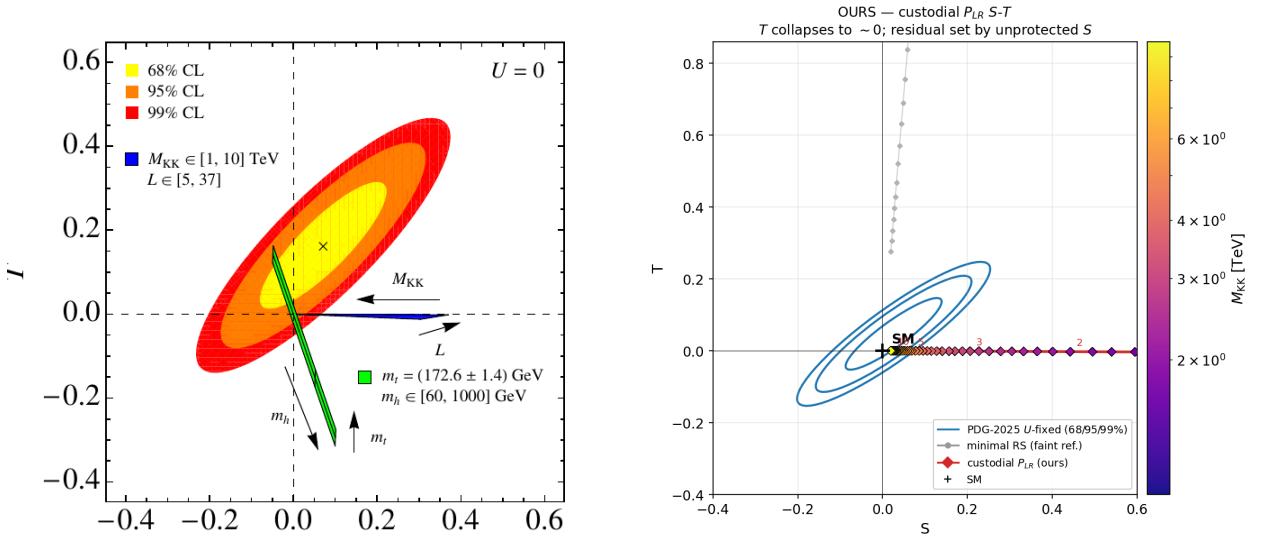


Figure 2: *Paper* (left): CGHNP 0807.4937 Fig. 4 **right panel**, the RS corrections *with* custodial protection. The blue region now runs flat along the S -axis at $T \approx 0$; the volume-enhanced T is gone. *Ours* (right): the custodial trajectory shows the same collapse, $T \rightarrow 0$, with the residual floor set by the *unprotected* S . Same axes, same behaviour — this paper panel is the right half of the very figure whose left half is the row above.

(b) $Z \rightarrow b\bar{b} — P_{LR}$ protects g_L^b at tree level

The minimal stripe below is also the $SU(2)_R$ -only result; only P_{LR} collapses it.

No custodial *and* $SU(2)_R$ only

paper (left) | ours (right)

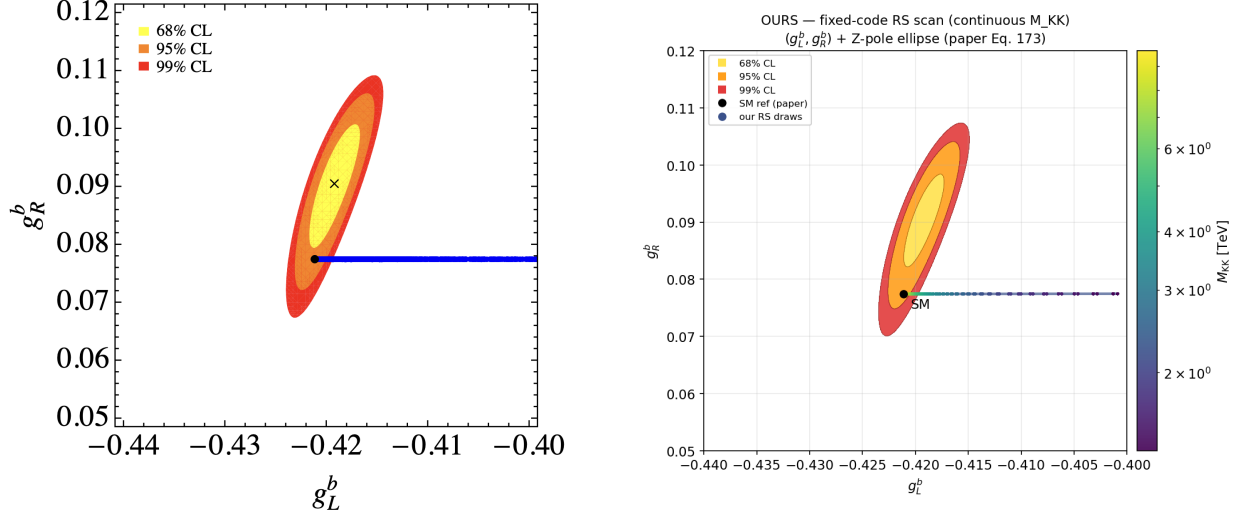


Figure 3: The (g_L^b, g_R^b) plane, same axes both sides. The minimal stripe sweeps away from the SM in g_L^b at nearly fixed g_R^b as M_{KK} drops. Our scan reproduces the paper stripe and SM point. $SU(2)_R$ -only sits on this same stripe: δg_L^b is unchanged from minimal (2.6×10^{-3}).

With P_{LR}

ours — no published (g_L^b, g_R^b) custodial panel exists

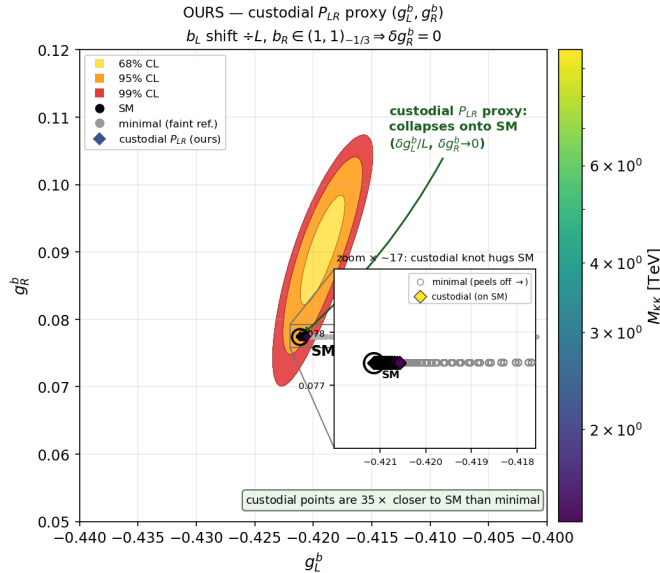


Figure 4: CGHNP plots only the minimal $Z \rightarrow b\bar{b}$ (Fig. 8 is minimal in both panels), so there is no published (g_L^b, g_R^b) figure for the custodial case to put on the left. *Ours*: with P_{LR} the b couplings collapse onto the SM ($\delta g_L^b \rightarrow 0, \delta g_R^b \rightarrow 0$ for $b_R \in (1, 1)_{-1/3}$), landing $\sim 35\times$ closer to the SM than the minimal/ $SU(2)_R$ stripe above. Top-partner loops partially re-introduce δg_L^b (CPSW).

(c) ε_K — custodial does *not* move the wall

No custodial = with custodial

paper (left) | ours (right)

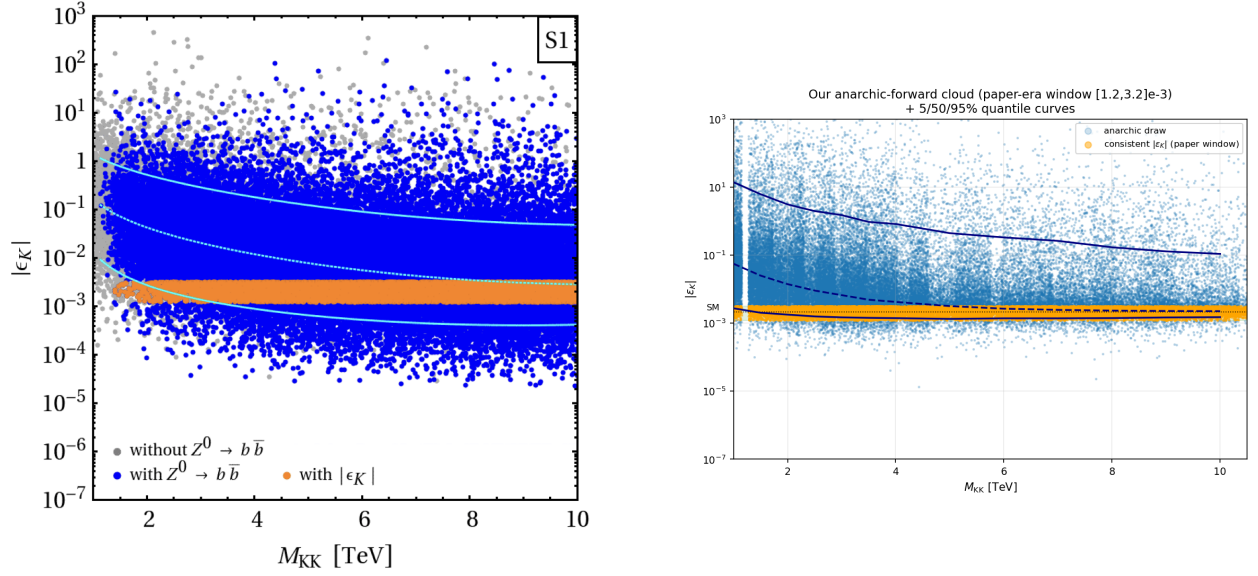


Figure 5: Anarchic $|\varepsilon_K|$ vs. M_{KK} , same axes both sides; both span ~ 6 decades with the 95%-quantile floor near 10 TeV. **This one row stands for both conditions:** custodial is blind to ε_K , because the binding operator is the color-octet KK-gluon C_4 , an electroweak singlet the custodial group cannot touch, so the cloud is *identical* with or without it. The full custodial calculation agrees: Blanke 0809.1073 computes $\Delta F = 2$ inside the custodial model and still finds $M_{KK} \gtrsim 18\text{--}30$ TeV.

4. Which papers we followed

Followed (formulas taken and coded up): the four electroweak-custodial papers. **CGHNP** (Casagrande et al., 0807.4937): the custodial T (Eq. 153), the minimal/custodial oblique, and the $Zb\bar{b}$ tree couplings (Eq. 170). **ADMS** (Agashe-Delgado-May-Sundrum, hep-ph/0308036): the custodial- T mechanism. **ACDP** (Agashe-Contino-Da Rold-Pomarol, hep-ph/0605341): the P_{LR} protection of Zb_L and the g_R embedding. **CPSW** (Carena-Pontón-Santiago-Wagner, hep-ph/0607106 and 0701055): the top-partner one loop.

Did not follow (cross-checks only, nothing implemented): the custodial-flavor papers. **CFW** (0804.1954) and **Blanke et al.** (0809.1073) and **Bauer II** (0912.1625), including Bauer II Fig. 13 / Eq. 131 (the custodial right-handed Z FCNC, enhanced $\sim 10\times$), are used only to confirm that the ε_K wall stays at tens of TeV. The split is electroweak vs. flavor: we follow the papers that give the custodial effect on S, T, U and $Zb\bar{b}$, the floors we run into, and we do not implement the custodial flavor effects, which stay subleading to the KK-gluon for ε_K . The detailed per-paper map is `docs/CUSTODIAL_PAPER_MAP.pdf`.

Bottom line

Custodial symmetry does exactly two things to what we run, and they are separable: $SU(2)_R$ protects the oblique T parameter (the S, T, U floor drops from 16 to 5.8 TeV, the residual set by the unprotected S), and the extra P_{LR} protects the $Zb_L\bar{b}_L$ vertex at tree level (top-partner loops partially undo it). It does **nothing** to ε_K : the binding kaon operator is the color-octet KK-gluon C_4 , an electroweak singlet, so the wall stays at tens of TeV — in our run and in Blanke’s full custodial calculation alike. Custodial is an electroweak cure, not a flavor cure. Only flavor *alignment* (RH-down $U(3)$ degeneracy, 5D-MFV, V_{5KM}) moves the ε_K wall.